

AppDev-1
PROJECT REPORT
Placement Portal Application

1. Student Details

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About Me:

I am a business professional with over a decade of leadership experience in advertising and business management. I have a strong interest in full-stack web development and machine learning. I enjoy building real-world applications that combine backend logic, data handling, and user-friendly interfaces.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To build a Placement Portal Application web application that allows Admin (Institute), Company, and students to interact with the system based on their roles. The system should allow students to apply for jobs, companies to manage recruitment drives, and administrators to oversee the entire placement process.

Approach:

The application was developed using Flask with a modular structure. The system supports role-based access for Admin, Company, and Student. It includes job postings, application tracking, placement management, and visualization of placement statistics.

Key design principles:

- Role-based access control
 - Relational database design
 - Clean UI with minimal CSS
 - Real-time status tracking
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3. AI/LLM Declaration

I used ChatGPT (GPT-5) to assist in:

- Debugging authentication and session handling
- Improving UI layout and user experience

The extent of AI usage is approximately 15-20%, mainly for guidance and debugging. All integration, testing, and final implementation were done manually.

4. Technologies and Frameworks Used

Technology Purpose

Flask	Backend web framework
SQLAlchemy	ORM for SQLite database
Flask-Login	Authentication and session management
Jinja2	HTML templating
Chart.js	Data visualization
SQLite	Database
HTML/CSS	Frontend UI

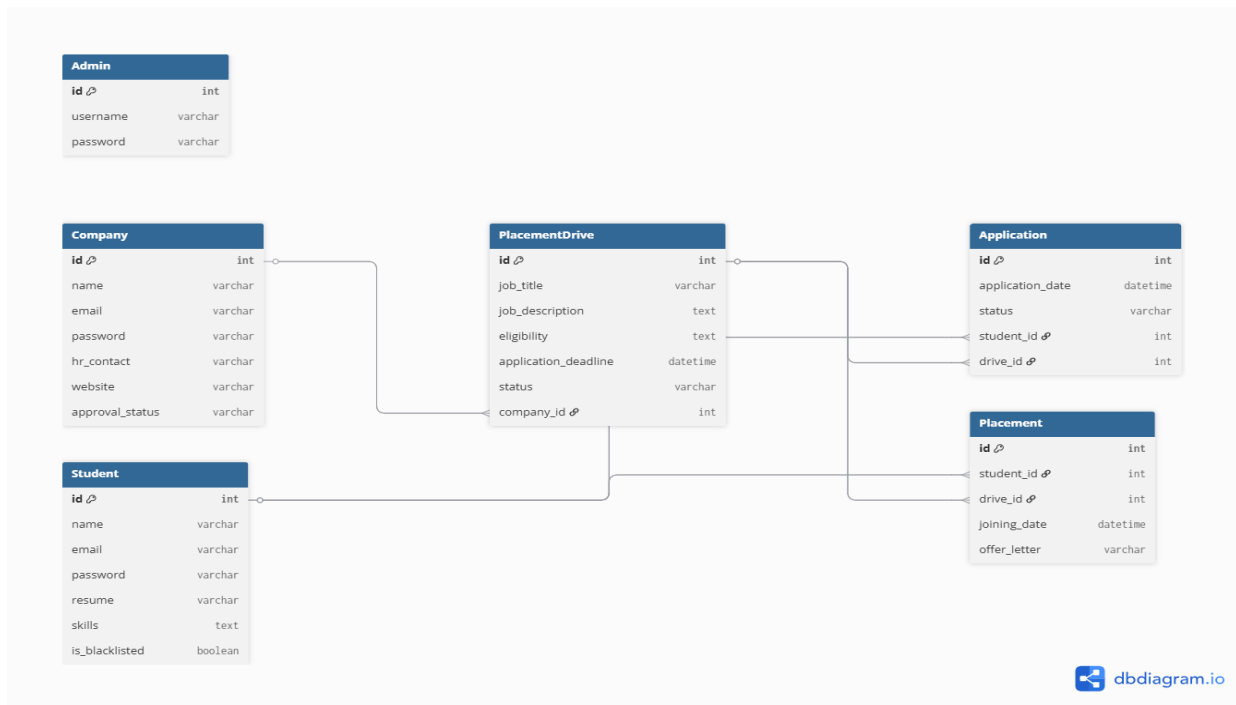
5. Database Schema / ER Diagram

Tables:

- **Admin** — stores admin credentials
 - **Company** — company profiles and approval status
 - **Student** — student profiles and resume data
 - **PlacementDrive** — job postings
 - **Application** — student applications to drives
 - **Placement** — final placement records
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Relationships:

- One-to-Many → Company → PlacementDrive
- One-to-Many → Student → Application
- One-to-Many → PlacementDrive → Application
- One-to-Many → Student → Placement
- One-to-Many → PlacementDrive → Placement



6. API Resource Endpoints

Endpoint	Method	Description
/login	POST	User login
/student/apply/<id>	GET	Apply for a job
/company/drive/create	POST	Create job posting
/company/application/<id>/update/<status>	GET	Update application status
/admin/company/<id>/approve	GET	Approve company
/admin/drive/<id>/approve	GET	Approve job posting
/student/chart-data	GET	Student analytics
/company/chart-data	GET	Company analytics

7. Architecture and Features

Architecture

- app.py → Main Flask application
- models.py → Database models
- templates/ → UI pages
- static/ → CSS and JS

Implemented Features

- ✓ Role-based authentication (Admin, Company, Student)
 - ✓ Company approval system
 - ✓ Placement drive creation and approval
 - ✓ Student job application system
 - ✓ Application status tracking (Applied → Placed)
 - ✓ Placement record management
 - ✓ Chart visualization using Chart.js
 - ✓ Search functionality (Admin)
 - ✓ Blacklisting system
 - ✓ Profile management (Student & Company)
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8. Video Presentation

Drive Link:

[\(Click Here!\)](#)